SSMRV College®

Department of Computer Applications

QUESTION BANK EVEN SEM-2026

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UNIT – 1
2 Mark Questions

1. Define random experiment with example.
2. What is Sample Space? Give example.
3. Define mutually exclusive events.
4. What are equally likely events?
5. Define Conditional Probability.
6. Define independent events.
7. State Addition theorem for two events.
8. State Multiplication theorem for two events.
9. What is Bayes' theorem? (Statement only)
10. Define an Event and Outcome.
11. A die is thrown once. Find the probability of getting a number greater than 4.
12. A card is drawn from a standard deck of 52 cards. Find the probability of drawing a king.
13. A coin is tossed once. Find the probability of getting a tail.
14. A number is selected from 1 to 10. Find the probability that the number is even.
15. A die is thrown. Find the probability of getting a multiple of 3.
16. From a pack of 52 cards, find the probability of drawing a red card.
17. Two coins are tossed. Find the probability of getting exactly one head.
18. A bag contains 3 red and 2 blue balls. One ball is drawn at random. Find the probability that it is blue.
19. A die is rolled. Find the probability of getting an odd number.
20. A student is selected randomly from a class of 20 students (12 boys and 8 girls). Find the probability of selecting a girl.
21. Two Coins are tossed. What is the Probability of getting both heads?
22. From a deck of 52 cards, two cards are drawn with replacement. Find the Probability that both the cards that are drawn are aces.
23. Three coins are tossed. Find the Probability that all are heads.
24. Three dice are thrown. Find the probability that all are sixes.
25. Three cards are drawn without replacement. Find the probability that all are kings.
26. A Bag has 5 red,3 blue balls. Three balls are taken without replacement. Find the Probability that all are red.

5 Mark Questions

1. A die is thrown. Find probability of getting an even number or a number divisible by 3.
2. From a pack of 52 cards, find probability of drawing a king or a red card.
3. Two coins are tossed. Find probability of getting at least one head.

4. A bag contains 5 red and 3 blue balls. Two balls are drawn without replacement. Find probability both are red.
5. Explain conditional probability with example.
6. Explain independent events with example.
7. Solve a problem using multiplication theorem.
8. Explain Bayes theorem with real-life example.
9. Two dice are thrown simultaneously, and the sum of the numbers obtained is found to be 7. What is the probability that the number 3 has appeared atleast once?
10. In a group of 100 computer buyers, 40 bought CPU, 30 purchased monitor and 20 purchased CPU and monitor. If a computer buyer chose at random and bought a CPU, what is the probability they also bought a monitor?
11. A class in advanced physics is composed of 10 juniors, 30 seniors and 10 graduate students. The final grades shows that 3 of the juniors, 10 of the seniors and 5 of the graduate students received an A grade for the course. If a student is chosen at random from this class and is found to have earned an A grade, what is the probability that he or she is a senior?
12. Find the probability of drawing a King or Queen.
13. Find probability of drawing an Ace or Spade.
14. Two cards drawn without replacement. What is the Probability of both are aces?
15. A die is thrown. Find probability of getting an even number or a multiple of 3.
16. Two cards drawn without replacement. What is the Probability that both are kings.
17. From a deck of 52 cards, find the probability of drawing a heart or a king.
18. Two dice are thrown. Find the Probability that both show 6.
19. A die is rolled. Find the probability of getting a number < 4 or even.
20. From the set of integers 1–15, Find the probability that the number is prime or multiple of 3.
21. From digits 1–9, two numbers are selected without replacement. Find the Probability that both the numbers that are selected is odd.
22. In a class of 100 students, 50 play Cricket; 40 play Football; 30 play Volleyball; 20 play Cricket & Football; 15 play Football & Volleyball; 10 play Cricket & Volleyball; 5 play all three. Find the probability that a student plays at least one sport.
23. Out of 200 students: 90 study Python; 80 study Java; 70 study Data Science; 30 study Python & Java; 25 study Java & DS; 20 study Python & DS; 10 study all. Find the probability that student studies at least one course.
24. From a Survey of 90 customers: 50 buy groceries; 45 electronics; 30 clothing; overlaps: 20,15,10; all: 5. Find the Probability that a customer will buy in atleast one category.
25. Out of 200 viewers: Netflix 120; Prime 100; Hotstar 80; overlaps: 50,40,30; all: 20. Find the Probability that a viewer will view atleast one OTT channel.
26. Out of 150 students attending seminars A, B, C: Counts: 80,70,60 ; Pairs: 30,25,20; Triple: 10. Find the Probabiity that a Student will attend atleast one seminar.

27. A shop buys products from two suppliers.

Supplier	Supply	Defect Rate
A	65%	2%
B	35%	5%

If a defective product is found, find probability it came from **Supplier B**.

10 Mark Questions

1. State and explain addition theorem for three events. Solve a suitable problem.
2. State and explain Bayes' theorem. Solve a real-life problem related to defective items.
3. A random sample of 200 adults are classified below by gender and their level of education attained:

Education	Male	Female
Elementary	38	45
Secondary	28	50
College	22	17

If a person is picked at random from this group, find the probability that
a)The person is a male, given that the person has a secondary education.
b)The person does not have a college degree, given that the person is a female.

4. A number is chosen from 1–20. Find probability it is divisible by 2 or 5.
5. In a factory, Machine A produces 60% of the items, and Machine B produces 40% of the items. 3% of the items produced by Machine A are defective, while 4% of the items produced by Machine B are defective. If an item is chosen at random and is found to be defective, what is the probability that it was produced by Machine A?
6. In a bag of marbles, 30% are red, 40% are blue, and 30% are green. Suppose that for these marbles: A red marble is shiny with a probability of 0.2; A blue marble is shiny with a probability of 0.3; A green marble is shiny with a probability of 0.4. If a shiny marble is picked at random, what is the probability it's red?
7. In a factory, Machine A produces 50% of the items; Machine B produces 30% of the items; Machine C produces 20% of the items. Defect rates: 2% items

from Machine A are defective; 3% items from Machine B are defective; 5% items from Machine C are defective. If a product is selected randomly and found defective, what is the probability it was produced by Machine B?

UNIT – 2
2 Mark Questions

1. Define random variable.
2. What is discrete random variable?
3. What is continuous random variable?
4. Define probability mass function (PMF).
5. Define probability density function (PDF).
6. Define cumulative distribution function (CDF).
7. What is mathematical expectation?
8. Define variance of a random variable.
9. What is function of a random variable?
10. Give example of continuous random variable.
11. A die is rolled. Let X be the number obtained. Write possible values of X.
12. Two coins are tossed. Let X be the number of heads. List the values of X.
13. Identify whether the following is discrete or continuous: Number of customers in a shop per hour.
14. Identify whether the following is discrete or continuous: Height of students.
15. If $P(X=1)=0.3$, $P(X=2)=0.5$, find $P(X=3)$.
16. Check whether the following is a valid PMF: $P(X=1)=0.2$, $P(X=2)=0.3$, $P(X=3)=0.6$
17. Find $E(X)$ if X takes value 2 with probability 1.
18. A fair die is rolled. Find $P(X=4)$.
19. If $E(X)=5$ and $E(X^2)=30$, find variance.
20. Write the formula for mean of discrete random variable.
21. If $P(X=0)=0.4$, $P(X=1)=0.6$, find $E(X)$.
22. Find $E(X)$ if X takes values 1,2 with equal probability.
23. If PDF is $f(x)=k$ for $0<x<1$, find k.
24. Find $P(0<X<1)$ if $f(x)=2x$.
25. If $Var(X)=4$, find standard deviation.

5 Mark Questions

1. Define discrete and continuous random variables with examples.
2. Explain PMF and its properties.
3. Explain PDF and its properties.
4. Construct probability distribution for number of heads in 2-coin tosses and find mean.
5. A die is thrown. Find mean of random variable X.
6. Given:

X	1	2	3
P(X)	0.2	0.5	0.3

Find the mean.

7. Verify whether given function is PMF and find missing probability:

X	**0**	**1**	**2**
P(X)	0.2	k	0.5

8. Find variance for:

X	**0**	**1**	**2**
P(X)	0.3	0.4	0.3

9. If Y=2X, find E(Y) when E(X)=3.
10. Find mean of continuous random variable:
    $f(x)=2x, 0<x<1$
11. Find k if $f(x)=kx, 0<x<2$
12. Find the expectation of:

X	**2**	**4**	**6**
P(X)	0.2	0.5	0.3

13. Find mean and variance for a fair coin toss (X=0 or 1).
14. Define a random variable. Explain with a suitable example.
15. Distinguish between discrete and continuous random variables with examples.
16. Explain Cumulative Distribution Function (CDF) and its properties.
17. Define Mathematical Expectation of a random variable. Give example.
18. Explain variance of a random variable and its importance.
19. What is a function of a random variable? Explain with example.
20. Explain the concept of expectation of a function of a random variable.
21. Write short notes on:
    (a) Mean
    (b) Variance
    (c) Standard deviation of a random variable

10 Mark Questions

1. Construct probability distribution for number of heads in 3 coin tosses.
   Find mean and variance.
2. A die is thrown twice. Let X be the sum. Construct distribution and find mean.

3. Given PMF:

X	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

   Find mean and variance.

4. A bag contains 3 red and 2 blue balls. Two balls are drawn. Let X = number of red balls. Find distribution and mean.
5. If PDF is: $f(x)=kx,\ 0<x<2$, Find: (a) k (b) mean (c) variance
6. Find mean and variance for: $f(x)=3x^2,\ 0<x<1$.

7. If $Y=X^2$, find distribution of Y when X = {1,2,3} with equal probability.

8. Given:

X	1	2	3	4
P(X)	0.1	0.2	0.3	0.4

   Find mean, $E(X^2)$, variance.

9. Construct PMF for number of heads in 2 coin tosses and find variance.

10. Find mean and variance for random variable:

X	-1	0	1
P(X)	0.2	0.5	0.3

11. Define random variable. Explain in detail types of random variables (discrete and continuous) with examples and differences.

12. Define and explain Probability Mass Function (PMF). Also discuss its properties with suitable example.

13. Define and explain Probability Density Function (PDF). Explain its properties and applications.

14. Explain Cumulative Distribution Function (CDF) in detail. Discuss its properties and relationship with PMF/PDF.

15. Explain Mathematical Expectation in detail. Discuss its properties and applications in real-life problems.

16. Define variance of a random variable. Explain its properties and significance in data analysis.

17. Explain the concept of function of a random variable in detail with real-life examples.

18. Derive and explain the formula for mean and variance of a discrete random variable.

19. Explain the concept of continuous random variable and discuss how mean and variance are calculated using PDF.
20. Write an essay on applications of random variables in real-world scenarios such as business, IT, healthcare, and finance.